NEPHROLOGY TEACHING SERIES

Cardiorenal Syndrome

Decongestion, not cosmetic creatinine watching

Cr rise during decongestion is usually hemodynamic — under-diuresing is the bigger danger.

Dr. Cheng • Premier Nephrology Medical Group

MS3/MS4 rotation teaching

Why we get consulted

Rising Cr during diuresis — 'should we back off?'

CONSULT TRIGGERS

- Rising Cr during HF decongestion
- Diuretic resistance
- Hypotension limiting GDMT
- Refractory volume overload
- KRT for volume (rare)
- Ultrafiltration request
- Balancing RAAS / SGLT2i in CKD+HF

TEACHING PEARL

Cr bump during active decongestion (weight down, BNP down, hct up) is benign hemodynamic. Under-diuresis is worse than a 20–30% Cr rise.

ACC/AHA 2022 HF guideline. UpToDate: Cardiorenal syndrome (2026).

Ronco classification (Types 1–5)

Useful framework but don't get hung up — Type 1 and 2 are where the action is.

Type	Name	Scenario
1	Acute cardiorenal	Acute decompensated HF → AKI. The inpatient consult.
2	Chronic cardiorenal	Chronic HF → CKD. The outpatient problem.
3	Acute renocardiac	AKI → cardiac dysfunction (volume overload, K+, acidosis → arrhythmia).
4	Chronic renocardiac	CKD → CV disease (accelerated atherosclerosis, LVH, CKD-MBD).
5	Secondary	Systemic process hits both (sepsis, amyloid, SLE, DM, sarcoid).

Ronco et al. JACC 2008.

Why the kidney fails in HF

It's not just low forward flow — venous congestion is the bigger driver.

VENOUS CONGESTION (main driver)

Elevated CVP → ↑ renal venous pressure → ↓ net filtration pressure. Strong correlation with AKI in acute decompensated heart failure (ADHF). Decongestion restores GFR even if cardiac output unchanged.

LOW CARDIAC OUTPUT

Falls primary role. Most HF patients presenting with AKI have normal or near-normal CO. Relevant in cardiogenic shock.

NEUROHORMONAL ACTIVATION

RAAS, sympathetic, vasopressin → Na/water retention + efferent vasoconstriction. ACEi/ARB/ARNi improve outcomes; a small hemodynamic Cr rise can be acceptable.

INFLAMMATION + DRUGS

Pro-inflammatory cytokines; NSAIDs, diuretic over-dose, contrast, sepsis overlay the primary problem.

Mullens et al. JACC 2009. UpToDate: Cardiorenal syndrome.

Decongestion — the strategy that works

Loop first, sequential nephron blockade if stuck. Watch markers, not just Cr.

STEP-UP DIURETICS

IV loop at 2.5× home oral dose (DOSE)

Furosemide 40–80 mg IV q12h or continuous infusion

Target: ≥ 3 L/24 h UOP; weight down 1–2 kg/day

If UOP < 100 mL in 2 h → double dose

Stuck? Add thiazide (metolazone 2.5–10 mg or HCTZ / chlorothiazide) — CLOROTIC: faster decongestion

Add acetazolamide 500 mg IV daily (ADVOR) — augments natriuresis, corrects bicarb rise

Last: ultrafiltration (NOT first-line; CARRESS-HF)

MARKERS OF EFFECTIVE DECONGESTION

Daily weight ↓ 0.5–1.5 kg

Rising hematocrit (hemoconcentration = good)

BNP / NT-proBNP trending down

Improving pulmonary exam, saturations, orthopnea

JVP lower; reducing edema

Urine Na > 50 mEq/L within 2 h of loop (good natriuretic response)

Modest Cr rise OK if all above trending right

DOSE (NEJM 2011). CLOROTIC (JAMA Intern Med 2023). ACC/AHA 2022 HF.

Creatinine up — what does it mean?

Context decides whether it's good, neutral, or a problem.

Scenario	Interpretation	Action
Cr ↑ 10–30% during active decongestion with weight loss + BNP down	*Hemodynamic; often protective*	CONTINUE diuresis; do NOT hold ACEi/ARB
Cr ↑ with weight UP, worsening exam, rising BNP	*Inadequate decongestion → worse perfusion*	ESCALATE diuretics (sequential blockade)
Cr ↑ with hypotension, oliguria, lactate up	*Low output / shock*	Reduce diuresis; consider inotropes, cath lab
Cr ↑ > 30%, abrupt, bland UA	*Over-diuresis / prerenal*	Pause loop, gentle albumin or NS, reassess volume
Cr ↑ + active sediment / proteinuria	*Intrinsic AKI — search*	Workup: ATN, GN, nephrotoxins, contrast

Mullens et al. JACC 2009. Damman et al. JACC 2013.

GDMT in CKD — keep it on

Patients with HF and CKD benefit MORE from GDMT than those without CKD.

Agent	When to continue / start	Stop / hold
ACEi / ARB / angiotensin-neprilysin inhibitor (ARNi)	Continue through 20–30% Cr rise; essential in HFrEF + CKD	K+ > 5.5 after binder, symptomatic hypotension, pregnancy
Beta-blocker (HFrEF)	Continue through decongestion; down-titrate if shock	Severe bradycardia; cardiogenic shock
MRA (spiro / epler) — HFrEF / heart failure with preserved ejection fraction (HFpEF)	eGFR ≥ 30, K+ ≤ 5.0. Monitor q1–2 wk after start	K+ > 5.5; severe CKD without binder
SGLT2i	Start at eGFR ≥ 20 (DAPA-HF, EMPEROR); continue to KRT	Acute illness, DKA risk in T1DM
Finerenone (DKD)	T2DM + CKD; additive to ACEi/ARB + SGLT2i	K+ > 5.0 at start; adjust per eGFR
Loop diuretic	Titrate to euvolemia; pre-discharge regimen PO equivalent	Over-diuresed / volume-depleted

DAPA-HF, EMPEROR-Reduced/Preserved, FIDELIO-DKD, FIGARO-DKD, PARADIGM-HF.

Landmark trials shaping cardiorenal practice

What to do when Cr rises during HF treatment.

Trial	Year	Takeaway
DOSE	2011	Low-dose vs high-dose loops, bolus vs continuous — high-dose = better decongestion, similar AKI.
CARRESS-HF	2012	Ultrafiltration NOT superior to stepped diuretics; more AEs. Diuretics first-line.
CLOROTIC	2023	Add HCTZ to IV loop → faster decongestion and weight loss in ADHF.
ADVOR	2022	Add IV acetazolamide to loop → greater natriuresis and decongestion.
DAPA-HF	2019	Dapagliflozin ↓ CV death/HF hosp 26% in HFrEF regardless of DM.
EMPEROR-Reduced	2020	Empagliflozin ↓ CV death/HF hosp 25% in HFrEF.
EMPEROR-Preserved	2021	SGLT2i benefit extends to HFpEF.
FIDELIO / FIGARO	2020–21	Finerenone ↓ CV and kidney events in DKD.
PARADIGM-HF	2014	Sacubitril/valsartan > enalapril in HFrEF; benefit in CKD subgroups.

Full citations on references slide.

Common pitfalls

The biggest errors lead to under-decongestion and re-admission.

AVOID

- Stopping diuretics for 20% Cr rise
- Reflexively holding ACEi/ARB
- NS boluses in congestion
- UF before sequential blockade
- No K+ monitoring on MRA
- Skipping SGLT2i in HFrEF + CKD
- Missing prerenal / hypotension

DO INSTEAD

- Decongest if weight / BNP improving
- Accept Cr ↑ up to 30% on RAAS
- Assess JVP + POCUS IVC
- Escalate: loop → thiazide → acetazolamide
- K+ weekly on MRA; binder if > 5
- SGLT2i when eGFR ≥ 20 and stable
- POCUS / CVP to distinguish shock

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 66-year-old man with HFrEF (EF 25%) and CKD 3a admitted with ADHF. On furosemide 40 mg PO daily at home, now on 80 mg IV q12h. Day 3: weight down 4 kg, BNP down from 2800 → 1400, crackles improving, hematocrit up from 36 → 40. Cr rose 1.5 → 1.9 (27% increase). Team wants to hold lisinopril, stop furosemide, give albumin.

QUESTION: Is this acute kidney injury that needs volume? What do you recommend?

Clinical case — answer

Key teaching points from the case.

ANSWER

No — this is hemodynamic Cr rise during effective decongestion. Continue loop diuretic, continue lisinopril. Add acetazolamide or metolazone only if diuresis stalls.

TEACHING

Rising Cr during active decongestion (weight down, BNP down, rising hct, improving exam) is usually benign and often PROTECTIVE. Stopping diuretics and 'protecting the kidneys' with saline causes reaccumulation of congestion and worse outcomes. Accept Cr rise of 20-30% on ACEi/ARB — it reflects hemodynamic adaptation, not injury. When truly stuck: sequential nephron blockade — add thiazide (CLOROTIC) or acetazolamide (ADVOR). UF is NOT first-line (CARRESS-HF). Start SGLT2i in HFrEF + CKD: dapagliflozin 10 mg (DAPA-HF) or empagliflozin (EMPEROR-Reduced) — down to eGFR 20.

REFERENCES

Cardiorenal Syndrome

Guidelines

ACC/AHA/HFSA 2022 HF Guideline

KDIGO 2024 CKD Guideline

KDIGO 2022 Diabetes in CKD

ESC 2021/2023 HF Guidelines

Landmark Trials

DOSE — NEJM 2011

CARRESS-HF — NEJM 2012

CLOROTIC — JAMA IM 2023

ADVOR — NEJM 2022

DAPA-HF — NEJM 2019

EMPEROR-Reduced — NEJM 2020

EMPEROR-Preserved — NEJM 2021

PARADIGM-HF — NEJM 2014

FIDELIO / FIGARO — NEJM 2020–21

UpToDate (2026 access):

Cardiorenal syndrome: definition, prevalence, diagnosis • *Treatment of heart failure with reduced ejection fraction* • *Diuretic resistance in heart failure* • *Use of loop diuretics in adults*